

Package Insert

Product Name



Ganadexil ®Enrofloxacin ~~Oral Solution~~ 100mg/ml **Oral Solution**

Enrofloxacin

Name and Strength of Active Substance(s)

Enrofloxacin 100mg/ml

Product Description

Ganadexil Enrofloxacin is a limpid and transparent solution without particles in suspension observable by simple visual inspection.

Pharmacodynamic

Mode of action

Two enzymes essential in DNA replication and transcription, DNA gyrase and topoisomerase IV, have been identified as the molecular targets of fluoroquinolones. They modulate the topological state of DNA through cleaving and resealing reactions. Initially, both strands of the DNA double helix are cleaved. Then, a distant segment of DNA is passed through this break before the strands are resealed. Target inhibition is caused by non-covalent binding of fluoroquinolone molecules to an intermediate state in this sequence of reactions, in which DNA is cleaved, but both strands are retained covalently attached to the enzymes. Replication forks and translational complexes cannot proceed beyond such enzyme-DNA-fluoroquinolone complexes, and inhibition of DNA and mRNA synthesis triggers events resulting in a rapid, drug concentration-dependant killing of pathogenic bacteria.

Antibacterial spectrum

Enrofloxacin is active against many Gram-negative bacteria, against Gram-positive bacteria and *Mycoplasma* spp.

In vitro susceptibility has been shown in strains of (i) Gram-negative species such as *Escherichia coli*, *Pasteurella multocida* and *Avibacterium (Haemophilus) paragallinarum* and (ii) *Mycoplasma gallisepticum* and *Mycoplasma synoviae*.

Types and mechanisms of resistance

Resistance to fluoroquinolones has been reported to arise from five sources, (i) point mutations in the genes encoding for DNA gyrase and/or topoisomerase IV leading to alterations of the respective enzyme, (ii) alterations of drug permeability in Gram-negative bacteria, (iii) efflux mechanisms, (iv) plasmid mediated resistance and (v) gyrase protecting proteins. All mechanisms lead to a reduced susceptibility of the bacteria to fluoroquinolones. Cross-resistance within the fluoroquinolone class of antimicrobials is common.

Pharmacokinetics

Enrofloxacin administered via drinking water to poultry is rapidly and very well absorbed with a bioavailability of approx. 90 %. Maximum plasma concentrations of 2 mg/L are reached within 1.5 hours after a single bolus dose rate of 10 mg/kg body weight with a total systemic availability of 14.4 mg·hr/L . Enrofloxacin is eliminated from the body with a total body clearance of 10.3 ml/min·kg. If dosed as continuous drinking water medication (multiple dosing) steady-state concentrations of 0.5 mg (turkeys) to 0.8 mg (chicken) enrofloxacin per litre are achieved. A high mean volume of distribution (5 L/kg) indicated good tissue penetration of enrofloxacin. Concentrations in target tissues like lungs, liver, kidney, intestine and muscle tissue, exceed plasma concentrations by far. In poultry enrofloxacin is poorly metabolised to its active metabolite ciprofloxacin (approximately 5%). Enrofloxacin is eliminated from the body at a half-life of 6 hours. Protein binding in poultry is approximately 25%.

Environmental Properties

Not applied.

Indication

Treatment of the infections caused by the following bacteria susceptible to enrofloxacin:

Chickens

Mycoplasma gallisepticum,

Mycoplasma synoviae,

Avibacterium paragallinarum,

Pasteurella multocida,

Escherichia coli.

Turkey

Mycoplasma gallisepticum,

Mycoplasma synoviae,

Pasteurella multocida,

Escherichia coli.

Target Species

Poultry: chicken, turkey

Recommended Dosage and Mode of Administration

Oral route in drinking water.

Chickens and turkeys

10 mg enrofloxacin/kg bodyweight per day for 3-5 consecutive days.

Treatment for 3-5 consecutive days; for 5 consecutive days in mixed infections and chronic progressive forms. If no clinical improvement is achieved within 2-3 days, alternative antimicrobial therapy should be considered based on susceptibility testing.

Via the drinking water. Always make sure that the entire dose offered has been consumed. The medicated water should be made up fresh each day just before it is offered to the animals. The drinking water must be medicated throughout the treatment period, and no other water source should be available. Determine the bodyweight of the birds as accurately as possible in order to avoid under dosing.

Use only fresh pre-solutions, prepared every day before start of treatment. Pumping systems should be checked constantly to assure proper medication. Empty the water system and fill it with medicated water before starting the treatment.

Calculate the daily quantity (ml) of Ganadexil ®Enrofloxacin Oral Solution 100mg/ml required for treatment period as follows:

Total number of birds x Average body weight in kg x 0.1= Total volume (ml) per day

Ganadexil ®Enrofloxacin Oral Solution 100mg/ml may be put directly into the header tank or introduced via a water proportioner pump.

Mode of Administration

Oral route in drinking water.

Contraindications

Do not use for prophylaxis.

Do not use when resistance/ cross-resistance to (fluoro)quinolones is known to occur in the flock intended for treatment

Do not use in the case of known hypersensitivity to the active substance, other (fluoro)quinolones or to any of the excipients.

Warnings and Precautions

This product shall be prescribed by registered veterinary surgeon only.

Special warnings for each target species

Treatment of *Mycoplasma spp* infections may not eradicate the organism.

Special precautions for use in animals

Official and local antimicrobial policies should be taken into account when the product is used.

Fluoroquinolones should be reserved for the treatment of clinical conditions which have responded poorly, or are expected to respond poorly, to other classes of antimicrobials.

Since enrofloxacin was first authorized for use in poultry, there has been widespread reduction in susceptibility of *E.coli* to fluoroquinolones and emergence of resistant organisms. Resistance has also been reported in *Mycoplasma synovia* in the EU.

Whenever possible, fluoroquinolones should only be used based on susceptibility testing.

Use of the product deviating from the instructions given in the label or package insert may increase the prevalence of bacteria resistant to the fluoroquinolones and may decrease the effectiveness of treatment with other quinolones due to the potential for cross resistance.

Special precautions to be taken by the person administering the medicinal product to animals

Those with known hypersensitivity to (fluoro)quinolones should avoid contact with this product.
Avoid contact with skin and eyes.

Rinse any splashes from skin or eyes immediately with water.

Wash hands and exposed skin after use.

Do not eat, drink or smoke while using the product.

Interactions with Other Medicaments

In vitro, an antagonism was shown, when combining fluoroquinolones with bacteriostatic antimicrobial agents such as macrolides or tetracyclines and phenicols. The simultaneous application of substances containing aluminium or magnesium can impair the absorption of enrofloxacin.

Statement on Usage During Pregnancy and Lactation

Do not use in laying hens producing eggs for human consumption.

Do not administer to layer replacement birds within 14 days of coming into lay.

Adverse Effects/ Undesirable Effects

None known.

If you notice any serious effects or other effects not mentioned in this package leaflet, please inform your veterinary surgeon.

Overdose and Treatment

No adverse clinical symptoms were observed in chickens and turkeys treated respectively with doses up to 10 and 6 times higher than the therapy dose.

The use of fluoroquinolones during the growth phase combined with a marked and prolonged increase in the intake of drinking water, and hence active ingredient, possibly due to high temperatures, may potentially be associated with damage of the articular cartilage.

Withdrawal Period(s)

Chickens: Meat and offal: 7 days.

Turkeys: Meat and offal: 13 days.

Not authorized for use in birds producing eggs for human consumption.

Do not administer to layer replacement birds within 14 days of coming into lay.

Storage Conditions

Keep out of sight and reach of children. Store below 25°C.

Shelf Life

After First Opening Of Container : 3 months.

After reconstitution or dilution with drinking water : 24 hours.

Incompatibilities

None known.

Disposal

Medicines should not be disposed of via wastewater or household waste. Ask your veterinary surgeon how to dispose of medicines no longer required. These measures should help to protect the environment.

Dosage Forms

Oral solution

Packaging available

1 Liter.

For animal treatment only. To be supplied only on veterinary prescription.

Name and Address of Manufacturer

Industrial Veterinaria, S.A.

Esmeralda,19

E-08950 Esplugues de Llobregat (Barcelona) Spain.

Marketing Authorization Holder

Taseen Trading Sdn. Bhd.

No 8, Jalan Astana 1F/KU2,
Bandar Bukit Raja, 41050,
Klang, Selangor DE. Malaysia

Date of Revision of Package Insert

~~12th November 2018~~ April 12, 2019